

Part C: Design Explanation

1. How would you evaluate this system before shipping it (offline evaluation and/or online monitoring)?

Offline evaluation. Each layer is tested separately, because a correct final answer can hide a fault in an earlier stage (classification, retrieval, generation, eligibility).

- **Golden question set:** Human-labeled questions with expected category, document, section, and answer. It covers every document, reworded questions, multi-document questions, both readings of the ambiguous rebate/billing document, unclear and unsupported questions, and eligibility cases (complete, missing facts, each rule failing).
- **Classification:** Compare predicted and expected categories, and track which categories are confused most often (e.g. rebates vs. billing).
- **Retrieval:** Measure recall@4, meaning the expected section appears in the top four chunks.
- **Answer grounding:** Check that every claim is supported by a retrieved section, figures and dates are correct, sources are relevant, and the system admits when documents lack the answer. Automated similarity cannot catch reversed meanings (income "below" vs. "above" a threshold), so policy answers also get human review.
- **Eligibility:** Test separately and strictly. Missing facts never produce a decision, any failed condition returns not eligible with a \$0 rebate, no partial estimate follows a failure, and collected facts are not altered before checking.

Release gates. Zero eligibility hard-gate failures, all fixed eligibility cases pass, and classification accuracy, retrieval recall, and grounding meet team-agreed targets. Each run records classifier, answer-writer, and embedding model versions, document index version, thresholds, and prompts so comparisons stay fair.

Online monitoring. Track success and failure rates, selected category and path, classification confidence, retrieved chunk IDs and scores, frequency of fallback to all-document search, retries, latency, and token cost. Alert on rising errors, latency, cost, retries, or low retrieval scores. Logs exclude raw questions and personal eligibility facts. Reviewing real conversations would need a separate, approved, privacy-safe sampling process.

2. What breaks first at 10x the document volume or 100x the query volume, and what would you change?

10x documents (about 140 documents). The current design rebuilds the entire index whenever the collection changes, re-chunking and re-embedding everything. This is acceptable at 14 documents and wasteful at scale, so it breaks first.

Changes (after measuring startup time, chunk count, embedding time, rebuild time, and search quality):

- 1 Embed chunks in batches.
- 2 Rebuild only new, changed, or deleted documents.
- 3 Build the new index separately, verify it, then switch searches over, so a failed rebuild never removes the working index.
- 4 Revisit one-section-per-chunk only if retrieval metrics show long sections hurting results.

100x queries. A knowledge question makes three slow external calls (classification, embedding, answer generation). Under load these cause queueing, provider rate limiting, higher cost, and retries that amplify traffic. These are likely to fail before local similarity search does.

Changes (after a load test measuring throughput, p50/p99 latency, failure rate, per-call time, and cost per request):

- Timeouts, concurrency limits, and connection reuse
- Multiple application replicas behind a load balancer
- Rate limits, plus cost limits and alerts

Session state risk. Incomplete eligibility facts live in one replica's memory, so a follow-up routed to another replica loses them. Initially, use sticky routing with short automatic expiry. At larger scale, move session state to shared storage, after deciding retention period, access control, protection, and deletion of personal facts.

3. If SunGrid added a brand-new document category next month, what in your design would need to change vs. stay the same?

Example: SunGrid adds "Electric Vehicle Charging."

What changes

- 1 Add the category to the shared list of allowed categories.
- 2 Update the classifier's validated output schema to accept it.
- 3 Map the new documents to the category in the ingestion step (a document can belong to several categories).
- 4 Add golden questions covering the new category, filtering, cross-category questions, and confusion with existing topics (e.g. charger installation vs. technical installation, charger rebates vs. incentives).
- 5 Re-check the confidence threshold, since overlapping categories may lower classifier confidence.

What stays the same

- Chunking, embedding, and similarity search
- Answer generation and source citation
- The eligibility hard gate and temporary eligibility conversation
- The response shape

Categories are metadata, so the pipeline (classify, filtered search, grounded answer) is unchanged. The exception is a new *action* rather than new information. "Tell me about vehicle chargers" is a document question, while "Schedule my charger inspection" would need a new tool or workflow.

4. Name one non-obvious trade-off you made while building this, and why you made it.

Decision: Each ## section becomes one chunk, with no fixed-size windows and no overlap.

Why

- A rule and its conditions stay together. Fixed-size splitting could separate "Members qualify for the rebate..." from "...only when income is below the threshold," and retrieving only the first half would produce a misleading answer.
- Citations map cleanly to meaningful document headings.
- Behavior is simple to understand and debug, and search does not return near-duplicate overlapping chunks.
- The corpus is small and already well structured, and no evaluation evidence shows section length hurting retrieval.

Cost: Section lengths vary widely (e.g. 60 words vs. 900 words). A long section may cover several ideas, produce a less precise embedding, match broad queries, and send unnecessary text to the answer writer.

Revisit condition: If retrieval evaluation shows long sections reducing recall, introduce smaller overlapping chunks for those sections only.